Matrix sequences and Markov chains

Lecture and worked examples

Objectives

Study a sequence of column matrices defined by $U_{n+1} = AU_n + B$; model a random evolution with two or three states by a probabilistic graph and a transition matrix; compute the distribution after n steps; find an invariant distribution; study convergence in the two-state case.

Introductory activity

In a simplified model (fictitious data), the weather on a given day is either fine (S) or rainy (P), and depends only on the weather the day before: after a fine day, it is fine with probability 0.8; after a rainy day, it is fine with probability 0.4. Day 0 is fine. Let s_n and p_n be the probabilities that day n is fine or rainy.

Introductory activity (2)

1. Represent the situation by a diagram: two vertices S and P, an arrow for each possible change, carrying its probability.
2. Using a tree, compute s_1, p_1, then s_2, p_2.
3. Justify that $s_{n+1} = 0.8s_n + 0.4p_n$ and $p_{n+1} = 0.2s_n + 0.6p_n$.
4. Let $\pi_n = \begin{pmatrix} s_n & p_n \end{pmatrix}$ and $T = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix}$. Check that $\pi_{n+1} = \pi_n T$. Compute π_{10} with a calculator. Start again with a rainy day 0. What do you notice?

Introductory activity (3)

5. Find the distribution $\pi = \begin{pmatrix} a & b \end{pmatrix}$, with $a + b = 1$, that satisfies $\pi T = \pi$. Compare it with the observations of question 4.

Introductory activity (4)

Observing is not proving: the properties conjectured here are established in the course.

Matrix sequences

Let A be a square matrix of order p, B a column matrix and (U_n) a sequence of column matrices such that $U_{n+1} = AU_n + B$ for every n. If $I_p - A$ is invertible, the matrix $C = (I_p - A)^{-1}B$ satisfies $C = AC + B$, and for every natural number n:

$$U_n = A^n (U_0 - C) + C.$$

Matrix sequences

$(I_p - A)C = B$ can be written $C = AC + B$. Subtracting: $U_{n+1} - C = A(U_n - C)$. The sequence $V_n = U_n - C$ satisfies $V_{n+1} = AV_n$, so $V_n = A^n V_0$ (induction), and $U_n = A^n(U_0 - C) + C$.

Matrix sequences

If all the entries of A^n tend to 0, then U_n converges to C (entry by entry). This behaviour can be observed with a calculator; it is taken for granted when the statement says so.

Markov chains

A system is observed at times $0, 1, 2, \ldots$; at each time, it is in one of the states $1, \ldots, k$ ($k = 2$ or 3). Let X_n be the state at time n. We say that (X_n) is a (homogeneous) *Markov chain* if, given the present state X_n, the distribution of X_{n+1} does not depend on the earlier states $X_0, \ldots, X_{n-1}$ (Markov property), and if the probability of moving from state i to state j does not depend on n (homogeneity). We then write, for all i, j and n such that $P(X_n = i) \neq 0$,

$$P_{(X_n=i)}(X_{n+1} = j) = t_{ij}.$$

The numbers t_{ij} are the *transition probabilities*.

Markov chains

The chain is represented by a *probabilistic graph*: a weighted directed graph whose vertices are the states, with an arc from i to j, of weight t_{ij}, when $t_{ij} > 0$. The *transition matrix* is the square matrix $T = (t_{ij})$. Its entries are nonnegative and the sum of each row is 1.

The *distribution* at time n is the row matrix $\pi_n = \begin{pmatrix} P(X_n = 1) & \cdots & P(X_n = k) \end{pmatrix}$; π_0 is the initial distribution.

For every natural number n: $\pi_{n+1} = \pi_n T$ and $\pi_n = \pi_0 T^n$.

By the law of total probability, with the partition $(X_n = 1), \ldots, (X_n = k)$ (events of probability zero are left out, as they contribute 0):

$$P(X_{n+1} = j) = \sum_{i=1}^{k} P(X_n = i)\, t_{ij},$$

which is entry j of $\pi_n T$. The second formula follows by induction.

If $P(X_0 = i) = 1$, then for every state j, $P(X_n = j)$ is the entry in row i and column j of T^n. In other words, the probability of moving from state i to state j in n transitions is $(T^n)_{ij}$.

The initial distribution is the row π_0 with a 1 in position i and 0s elsewhere. Then $\pi_n = \pi_0 T^n$ is row i of T^n.

Markov chains

A distribution π (a row of nonnegative entries with sum 1) is *invariant* if $\pi T = \pi$. If $\pi_0 = \pi$, then $\pi_n = \pi$ for every n.

Markov chains

Let $T = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}$ with $0 \leq a \leq 1$, $0 \leq b \leq 1$ and $0 < a+b < 2$. The chain has a unique invariant distribution $\pi = \begin{pmatrix} \frac{b}{a+b} & \frac{a}{a+b} \end{pmatrix}$, and whatever the initial distribution, π_n converges to π.

Write $\pi_n = \begin{pmatrix} x_n & 1 - x_n \end{pmatrix}$. Then $x_{n+1} = (1-a)x_n + b(1-x_n) = (1-a-b)x_n + b$. This arithmetico-geometric sequence has fixed point $\ell = \dfrac{b}{a+b}$, and $x_n - \ell = (1-a-b)^n(x_0 - \ell)$. Since $-1 < 1-a-b < 1$, $(1-a-b)^n \to 0$ and $x_n \to \ell$. An invariant distribution satisfies $x = (1-a-b)x + b$, so $x = \ell$: it is unique.

Solve the system formed by $\pi T = \pi$ and the condition $a + b + c = 1$; one of the three equations of $\pi T = \pi$ follows from the other two. Possible convergence of π_n is observed with a calculator.

The distribution is a row and we multiply on the right by T: $\pi_{n+1} = \pi_n T$, not $T\pi_n$. It is the rows of T, not its columns, that add up to 1.